# Problem Set 1
# ECO3121 - Fall 2022

# September 23, 2022

**Due 3PM, 09/30/2022**
Please remember to submit your Stata code and requested output as it will be graded.

## Question 1

Download from the blackboard site and load into STATA the agricultural household data `aghousehold.dta`. The main data we use is the National Fixed Point Survey (NPFS), which is a nationally representative panel dataset (unbalanced) of roughly 5000 households in 88 villages between 1995 and 2002. It is collected by the Ministry of Agriculture and Rural Affairs of China.

Since this will be a univariate model, we'll be estimating regressions of the form:

$$yield_i = \alpha + \beta_1 fertilizer_i + \mu_i$$

First, let's analyze the effect of fertilizer application on yield.

1. Since we are learning simple linear regression and multiple linear regression at current stage, you can limit the sample to households in year 2002 by running code "keep if year==2002" in Stata. you can generate variable yield (output per unit of land) via $\frac{hx95_nh125}{hx95_nh112}$, and variable fertilizer application intensity (fertilizer application per unit of land) through $\frac{nh6_3}{hx95_nh112}$. (1 point)

2. What sign do you predict $\beta_1$ will have? Why? (1 point)

   *Answers: positive. Fertilizers contribute to yield growth.*

3. Run the regression and report the estimated coefficients, their standard errors, and $R^2$. Does the estimated value of $\beta_1$ agree with your prediction? (2 point)

   *Answers: Yes.* $\hat{\beta}_1 = 0.051$, $Se(\hat{\beta}_1) = 0.007$, $R^2 = 0.011$

| | (1) |
|---|---|
| Depend Var | Yield |
| Fertilizer | 0.051*** |
| | (0.007) |
| Constant | 352.757*** |
| | (2.574) |
| Observations | 4,076 |
| R-squared | 0.011 |

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

4. Interpret the value of the estimated parameters $\hat{\beta}_0$ and $\hat{\beta}_1$ from the model. (2 point)

   *Answers: An additional kilogram increase in fertilizer application intensity is associated with 0.051 kilogram increase in yield. Without application of fertilizers, the yield is 352.75 kg/mu.*

5. Compute the fitted value $\widehat{yield}_i$ and the residual $\hat{\mu}_i$ for each observation, and verify that the residuals (approximately) sum to 0. (1 point)

   *Answers: Yes. Sum=-.00044052* $\approx 0$. See the code.

6. According to the estimated relation, what is the predicted $\widehat{yield}_i$ for a farmer with a fertilizer application intensity of 100 kg/mu? (pay attention to units). (1 point)

   *Answers: Yes.* $\hat{y} = 0.05 * 100 + 352.75 =$ *357.75 kg/mu.*

7. How much of the variation in yields for these farmers is explained by their fertilizer input intensity? Can you provide some reasons for why yields might be lower among poorer farmers? (2 points)

   *Answers: 1.1%. Open question. E.g., Poor farmers are more resource constrained and then less productive*

8. An professor asks you whether they should estimate a linear relationship between yields and fertilizer input. You are thinking of introducing quadratic form in the regression. Explain why are you interested in quadratic regression form and report your results and interpretations (3 points)

   *The turning point is at* $-\frac{\beta_1}{2*\beta_2} = -0.132/(-0.000134*2) = 492.537$. *Yield increases with fertilizer application until about fertilizer intensity equal to 492.537 kg/mu and decreases thereafter.*

| VARIABLES | (1)<br>Yield |
|---|---|
| fertilizer | 0.132*** |
| | (0.012) |
| fertilizer_sq | -0.000134*** |
| | (1.57e-06) |
| Constant | 343.140*** |
| | (2.789) |
| Observations | 4,076 |
| R-squared | 0.029 |

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

9. The professor sees your analysis. She asks you to explore alternative functional forms for this relationship. Using the provided data, you estimate the following equations:
   * A log-linear relationship: $log(yield_i) = \beta_0 + \beta_1\ fertilizer_i + \mu_i$
   * A linear-log relationship: $yield_i = \beta_0 + \beta_1\ log(fertilizer)_i + \mu_i$
   * A log-log relationship: $log(yield)_i = \beta_0 + \beta_1\ log(fertilizer)_i + \mu_i$

   `We take natural log here. In Stata, the coding is "log(variable + 1)"`

   Run the regressions recommended by the professor. Interpret the parameters on fertilizer application intensity (or log(fertilizer application intensity)) in each of these equations. (3 points)

| VARIABLES | (1)<br>logYield | (2)<br>Yield | (3)<br>logYield |
|---|---|---|---|
| fertilizer | .0001188*** | | |
| | (.0000245) | | |
| logfertilizer | | 45.272*** | 0.168*** |
| | | (2.179) | (0.007) |
| Constant | 5.776*** | 162.150*** | 5.058*** |
| | (0.008) | (9.776) | (0.032) |
| Observations | 4,076 | 4,076 | 4,076 |
| R-squared | 0.006 | 0.096 | 0.122 |

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

*1) One kilogram increase in fertilizer application is associated with .012% increase in yield.*
*2) One percent increase in fertilizer application is associated with 0.45 kilograms increase in yield.*
*3) One percent increase in fertilizer application is associated with 0.168 percent increase in yield.*

10. Repeat *question 1* for households in year 2001 only (linear-linear regression). How do your estimates of $\widehat{\beta}_0$ and $\widehat{\beta}_1$ differ from *question 1*? Briefly explain the possible reasons in 3-5 sentences. (2 points)

| | (1) |
|---|---|
| VARIABLES | Yield |
| fertilizer | 0.140*** |
| | (0.015) |
| Constant | 336.232*** |
| | (2.981) |
| Observations | 4,200 |
| R-squared | 0.020 |

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

The rainfall conditions could be largely different/ fertilizer quality could be different across these two years.

11. Describe the SLR4 assumption under this case. Do you think that assumption holds true? (1 point)

*The independent variable fertilizer application must not contain information about the mean of the unobserved factors $\mu$.* ***Not true.***

## Question 2

A large-landholding farmer outsmarts others by running an experiment to measure the effect of fertilizer application on agricultural yields. She **randomly assigns** her 400 land plots into two groups. In the first group, she inputs 20 kg/mu of fertilizers, while the other group receives 40 kg/mu of fertilizers. She also keeps other agricultural inputs (water, pesticides, labors) equally between these two groups. Let $Y_i$ denote the yields for $i^{th}$ plot, let $X_i$ denote the amount of fertilizer application per unit of land, ($X_i = 20$ or 40), and consider the regression model $Y_i = \beta_0 + \beta_1 X_i + \mu_i$

1. Explain what the term $u_i$ represents. Proposal a potential reason that why different plots might have different values of $\mu_i$? (2 point)

*The random error represents everything that the regression model does not have into account. For instance, the land qualities are different in those 400 plots. This can cause $\mu_i$ to be different.*

2. Is $E[u_i|X_i] = 0$ in this case? Explain why or why not? (1 point)

   *Yes. Because the variable is randomly assigned.*

3. The estimated regression is $Y_i = 300 + 0.7X_i + \mu_i$.

   (a) Is $\beta_1 = 0.7$ an unbiased estimator? (1 point)

   *Yes. Because the variable X is randomly assigned. $E[u_i|X_i] = 0$.*

   (b) After you got this estimation, the farmer's assistant tells you that he is concerned that the program may not actually have randomly allocated fertilizer quantity across the 400 plots, and that some cheating may have gone on (he heard that the richest land (land with highest soil quality) were more likely to be put into the more fertilizer group). What concern would this give rise to in your estimation? (1 point)

   *We'll have selection bias. If land plots were selected based on their initial land quality ("richest lands"). we would expect that one year later the average yield in the more-fertilizer group is higher than the average yield in the low-fertilizer group because of the fact that land were richer one year prior and not because of more fertilizers. Therefore, our estimate of the treatments is biased upwards.*

   (c) What additional variable (data) would you like to collect to verify whether the assistant's concern is true, and what regression specification would you use these data for to investigate whether his concern is true? (3 points)

   We should ask the assistant to give us data on land quality for each plot before the experiment started one year prior ($Z_i^{t-1}$). By comparing the mean land quality of more-fertilizer group and low-fertilizer group we can directly see if more-fertilizer groups were richer in land before the experiment.

   In particular, we could run the following regression,

$$Y_i^{t-1} = c + \beta X_i + \epsilon_i$$

   If the X is randomly assigned the coefficient $\beta$ should be not statistically different from zero which implies that land quality in low-fertilizer group is not lower than in more-fertilizer group. If $\beta$ is larger than zero (and statistically different from zero) the concern the assistant raised is valid and concerning.

   or

$$X_i = c + \beta Y_i^{t-1} + \epsilon_i$$